PACE: Probabilistic Adaptive Contention Control for Non-Primary Channel Access in IEEE 802.11bn

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***Abstract*—While an overlapping basic service set (OBSS) transmission occupies the primary channel, adjacent non-primary channels sit idle because conventional IEEE 802.11 mandates that the primary channel be included in every access. IEEE 802.11bn introduces non-primary channel access (NPCA) to exploit these idle intervals. The standard contention mechanism, however, is ill-suited to NPCA's constrained transmission windows, as its collision-triggered backoff growth synchronizes stations into repeated collisions or pushes their counters beyond the remaining opportunity, wasting airtime that could carry data. We propose PACE, a probabilistic adaptive contention control scheme that lets stations track the time-varying fair-share transmission probability of the depleting window without central coordination. PACE combines a deadline-scaled initialization and a distributed multiplicative feedback rule that copies the successful sender's probability on solo receptions, together with a proactive self-exclusion mechanism that silences stations whose frame length exceeds the remaining window. We evaluate PACE under IEEE 802.11 standard timing on a non-primary channel shared with native stations, against the standard-compliant NPCA baseline and a genie-aided fair-share reference. Across 5–50 visiting stations, PACE keeps the total useful airtime within 3% of the fair-share reference under request-to-send/clear-to-send (RTS/CTS)-protected access, up to 61% above the standard procedure and up to $4.1\times$ under basic access, while holding the visitors' airtime at or below their population-proportional share. The advantage grows with the visitor population, where standard backoff idles behind frozen counters and collides in synchronized bursts. These results establish probabilistic, deadline-aware contention as a design reference for non-primary channel access in next-generation Wi-Fi.**

***Index Terms*—Non-Primary Channel Access, IEEE 802.11bn, Probabilistic Access Control, OBSS, Adaptive Contention, Wi-Fi**

I. Introduction

Spectrum efficiency in next-generation Wi-Fi networks depends critically on how effectively stations (STAs) exploit available channel time. In dense deployments, overlapping basic service sets (OBSSs) routinely occupy the primary channel while adjacent channels remain idle, creating a persistent underutilization problem. Hence, IEEE 802.11bn addresses this gap by introducing non-primary channel access (NPCA), which allows STAs to transmit opportunistically on non-primary channels during ongoing OBSS intervals [1], [2].

NPCA, however, presents a contention challenge distinct from conventional channel access. Each opportunity lasts only as long as the ongoing OBSS transmission, so competing STAs must resolve their access within a finite, depleting window. To govern this access, the IEEE 802.11bn draft has STAs on the non-primary channel reuse the same binary exponential backoff (BEB) mechanism that the distributed coordination function (DCF) employs on the primary channel [1], yet BEB was designed for open-ended contention. On the primary channel, doubling the backoff window after each collision harmlessly spreads out retransmissions; within a bounded NPCA window the same growth turns harmful, as the inflated backoff frequently exceeds the time remaining. STAs whose counters overshoot the deadline are silenced for the rest of the interval, leaving the channel idle even when they still have data to send. Heterogeneous frame lengths compound the problem, since the draft leaves unspecified what a STA should do when its frame no longer fits the remaining window; a compliant STA may keep contending, or transmit only to be cut off at timer expiry, despite having no viable transmission opportunity.

Existing work on NPCA has focused on characterizing aggregate throughput and delay under the standard switching rule, taking BEB within the non-primary channel as given [2]–[4]. The contention dynamics within the finite NPCA window, and the mismatch between BEB's infinite-horizon design and the window's bounded duration, have not been addressed. Adaptive probabilistic schemes, in turn, are built for infinite-horizon operation and converge to a static equilibrium, so they too fail to track the optimal access probability as the window shrinks and the set of viable STAs changes.

In this paper, we propose PACE, a probabilistic adaptive contention control scheme for NPCA that enables STAs to collectively track the time-varying fair-share transmission probability of each moment within the window, without central coordination or explicit inter-STA signaling. PACE adapts the multiplicative-increase and multiplicative-decrease (MIMD) feedback rule inspired by the probabilistic neighbor discovery (PND) algorithm [5] to the finite-window NPCA context. STAs copy the successful sender's probability on solo receptions, reduce on collision, and increase on idle slots, causing distributed dynamics to track the fair-share policy. PACE further adds a physical protocol data unit (PPDU)-aware self-exclusion rule that silences STAs whose frame length exceeds the remaining window, eliminating a class of futile

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contention attempts that BEB would otherwise sustain.

The design of PACE is motivated by two requirements that are unique to NPCA's bounded-window structure: 1) the fair-share transmission probability is time-varying and must track the depleting window rather than converge to a static equilibrium, and 2) STAs whose frame length exceeds the remaining window must proactively self-exclude to prevent futile contention that wastes viable slots. The main contributions of this paper are as follows.

- We characterize the fair-share transmission probability for finite-window NPCA, showing that it must continuously track the remaining window duration and the number of still-viable competing STAs, a deadline-driven, rising target that neither BEB nor existing infinite-horizon adaptive schemes can follow, since the latter react to a changing contending population but never to a window that is running out.
- We propose PACE, a fully distributed algorithm that addresses both requirements simultaneously. A deadline-scaled initialization and MIMD feedback drive the transmission probability toward the time-varying target without inter-STA signaling or knowledge of the total STA count.
- Through simulation in IEEE 802.11 standard timing on a channel shared with native DCF stations, we demonstrate that PACE keeps the total useful airtime within 3% of a genie-aided fair-share reference under request-to-send/clear-to-send (RTS/CTS)-protected access and up to $4.1\times$ above the standard-compliant baseline under basic access, holds the visitors' airtime at or below their population-proportional share, and attains a per-attempt success ratio close to three times the baseline's, matching the fair-share reference; an ablation study further shows that both the deadline-scaled initialization and the self-exclusion rule are necessary for these gains.

The remainder of this paper is organized as follows. Section II reviews related work on NPCA and adaptive medium access control (MAC) protocols. Section III presents the system model. Section IV describes the PACE algorithm and its theoretical foundation. Section V presents simulation results, and Section VI concludes the paper.

II. Related Work

This section reviews the two lines of work on which PACE builds, namely NPCA in IEEE 802.11bn and adaptive contention control in wireless LANs, and closes by contrasting them with PACE.

A. Non-Primary Channel Access in IEEE 802.11bn

IEEE 802.11bn introduces NPCA as a core mechanism for exploiting idle secondary channels during OBSS transmissions [1], [6]. Wei *et al.* [2] extended Bianchi's DCF model to a multi-channel setting, establishing analytically that NPCA always outperforms the non-NPCA mode; a follow-up study [4] incorporated switching overhead and proposed a threshold policy to toggle between NPCA and legacy modes based on primary channel occupancy. Bellalta *et al.* [3] developed a Continuous-Time Markov Chain model for a multi-BSS topology, identifying a secondary-channel contention trade-off that can limit NPCA gains when the secondary channel is shared. Lee and Song [7] further showed that PPDU frame duration has a nonlinear effect on inter-BSS coexistence, and Song [8] applied deep reinforcement learning to the NPCA channel-activation decision.

Across these studies, however, efforts to improve NPCA performance target the decision of when to switch to the non-primary channel, through occupancy-based threshold policies [4] or learned activation rules [8], while the contention mechanism exercised within the window is left fixed at BEB. How STAs should adapt their access probability as the window depletes and viable STAs drop out has not been studied.

B. Adaptive Contention Control in Wireless LANs

Bianchi [9] established that BEB operates below the theoretical throughput ceiling, and Cali *et al.* [10] showed that p-persistent access can approach this limit in a distributed manner, at the cost of estimating the number of active stations at run time from the observed idle-slot statistics. Subsequent work has diversified the adaptation mechanisms. BLADE [11] uses a hybrid increase multiplicative decrease (HIMD) policy driven by a universally observable channel signal, although the target access rate it regulates to is a fixed constant derived from a steady-state throughput model; Wilhelmi *et al.* [12] impose structured transmission order via inter-BSS activity awareness, which requires each STA to identify its neighbors and maintain a list of them to derive its position in the order; and learning-based approaches optimize the contention window through deep Q-networks [13], where the access point observes the collision rate and throughput of the whole network, selects a contention window threshold, and hands it down to its associated STAs in its beacons, or through fairness-constrained multi-agent policy optimization [14] trained centrally on the joint observations of all agents.

Several of these schemes do follow a moving operating point, tracking a target that shifts as the contending population changes. What none of them represents is a deadline. All of them assume STAs contend over an open-ended horizon, so their target never depends on how much time is left. NPCA's bounded window demands a fundamentally different approach. The optimal transmission probability must evolve as the window shrinks and the viable STA population changes. The MIMD feedback rule of PND [5] is the closest mechanism in spirit, but was designed for infinite-horizon neighbor discovery and provides no accounting for bounded window duration or frame-length viability.

Table I summarizes these distinctions along three axes. *Frame-fit self-exclusion* withdraws a STA whose frame no longer fits the remaining window. A *signaling-free* scheme transmits no control information for its own sake, a value riding in a frame a STA sends anyway not being counted, and $\triangle$ marks a scheme that adds no transmission but must still infer the contending population from the activity of the others. *Time-varying target tracking* follows an operating point

TABLE I: Comparison of contention control schemes.

Method	**Mechanism and objective**	**Frame-fit self-exclusion**	**Signaling-free**	**Time-varying target tracking**
PND [5]	Multiplicative feedback with solo-copy rule; converge the transmission probability to the optimum without the contender count	×	✓	✓
AND [15]	Fixed-probability slotted ALOHA; discover all neighbors in finite time without coordination	×	✓	×
Cali *et al.* [10]	p-persistent access with run-time estimation of the active station count; approach the theoretical throughput limit in a distributed manner	×	△	✓
BLADE [11]	Hybrid increase / multiplicative decrease toward a fixed target access rate; eliminate packet-delivery droughts for real-time Wi-Fi	×	✓	×
IYT [12]	Neighbor-ordered backoff interval assignment; reduce collision probability and worst-case latency	×	△	×
ACWB-DQN [13]	Deep Q-network contention window selection centralized at the AP; maximize throughput under varying network load	×	×	✓
FC-MAPPO [14]	Fairness-constrained multi-agent policy optimization; minimize collisions while ensuring inter-STA fairness	×	×	✓
PACE (proposed)	**Multiplicative feedback with PPDU-aware self-exclusion; maximize channel utilization in the finite NPCA window**	✓	✓	✓

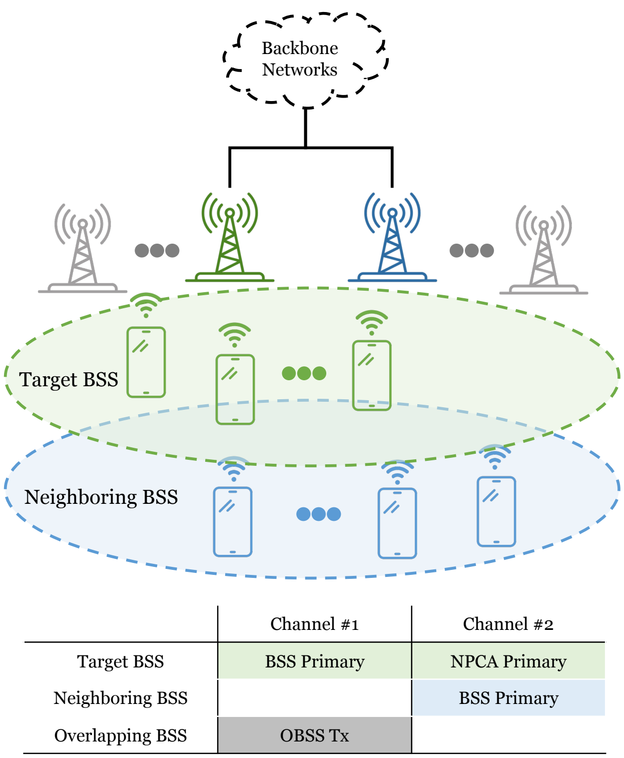


Fig. 1: Two-BSS topology and channel assignment.

that moves with the contending population rather than a static equilibrium. No prior scheme withdraws an unfittable frame, and the schemes that do adapt their operating point either track only the contending population or pay for the adaptation with a view of that population, whereas PACE meets all three requirements from purely local observations.

III. System Model

A. Network and Channel Model

Fig. 1 depicts the two-BSS topology and channel assignment assumed throughout this work. The target BSS, whose NPCA operation is our focus, comprises an access point (AP) and a set $\mathcal{V} = \{1, 2, \ldots, N_{\text{vis}}\}$ of associated non-AP STAs and they can be considered as *visitors* to the neighboring BSS. The visitors normally contend on their BSS primary channel, Channel #1 in Fig. 1, which is the main channel used by the target BSS for its own transmissions.

Channel #1 is also used by an OBSS operating on the same channel, whose transmissions intermittently render it busy. In standards prior to IEEE 802.11bn, a STA could not initiate a transmission whenever its designated primary channel was busy, even if the secondary channel was idle. In IEEE 802.11bn, however, the AP can designate a separate channel as the NPCA primary channel, used by the visitors during NPCA operation, named as Channel #2 in Fig. 1. We use the qualifier “BSS” to distinguish the original primary channel from the NPCA primary channel, consistent with P802.11bn Draft terminology [1].

The NPCA primary channel is not unused spectrum reserved for the target BSS. Channel #2 is itself the BSS primary channel of the neighboring BSS, comprising its own AP and a set $\mathcal{N} = \{1, 2, \ldots, N_{\text{nat}}\}$ of *native* STAs that contend on it with standard DCF at all times. The visitors, by contrast, use Channel #2 only opportunistically, during NPCA visits. Two classes of contenders therefore coexist on the NPCA primary channel: the native STAs, which contend there persistently, and the visitors, whose access is confined to the bounded NPCA window.

As depicted in Fig. 1, we assume the coverages of the two BSSs coincide, so that all visitors and native STAs are mutually within carrier-sensing range and form a single collision domain on the NPCA primary channel: every visitor senses, defers to, and competes with every native STA. This is the configuration in which the visitors’ impact on native airtime is greatest, so the coexistence results reported here upper-bound that cost. In general the two coverages may overlap only partially, in which case a visitor outside the neighboring BSS coverage would reuse the NPCA primary channel without contending with the natives at all; this spatial-reuse regime only lowers the coexistence cost and is left to future work.

B. NPCA Operation and Effective Window

Fig. 2 contrasts the medium-access processes with and without NPCA operation. Under conventional access (top),

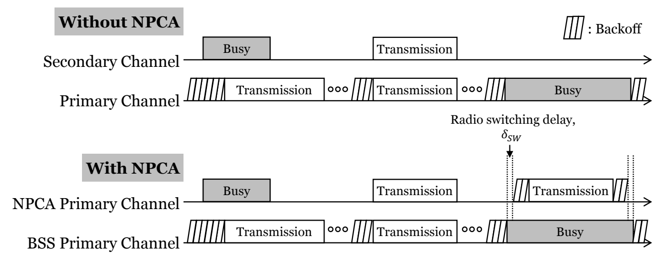


Fig. 2: The medium access process comparison with and without NPCA.

prior to IEEE 802.11bn, a STA contends only on the BSS primary channel.[1] It decrements its backoff counter while the channel is idle and transmits when the counter reaches zero. If the secondary channel is also sensed idle at this instant, the STA may bond it with the BSS primary channel and transmit a wideband PPDU spanning the combined bandwidth (e.g., 40, 80, 160, or 320 MHz). When an OBSS transmission seizes the BSS primary channel, however, the STA must defer and freeze its backoff for the entire OBSS busy period, even though the secondary channel is idle.

Under NPCA access (bottom), a STA no longer wastes the OBSS busy period. When it detects an OBSS transmission on the BSS primary channel, it decodes the preamble of that transmission and learns its duration in advance,[2] rather than freezing its backoff for the whole period, it switches to the idle NPCA primary channel and contends and transmits there. On each such switch the STA saves its backoff state on the BSS primary channel and initializes a fresh contention window for the NPCA primary channel, so every NPCA visit begins at the minimum backoff stage regardless of prior history. The STA already knows when the OBSS transmission will finish, so it sets a timer to that instant and switches back to the BSS primary channel when the timer expires, restoring the saved state without having to re-sense the channel. In this way the secondary spectrum that the a conventional STA leaves idle is put to use, and a STA can complete one or more transmissions that it would otherwise have had to postpone.

Two constraints shape this access. First, a radio transition delay δ_{sw} is incurred on each switch: the STA becomes ready on the NPCA primary channel only after the switching delay, and the NPCA timer is set to expire one switch-back delay before the OBSS transmission ends so that the STA is back on the BSS primary channel in time [1]. Second, the opportunity lasts only until the known end of the OBSS transmission, whose remaining duration D_{rem} is read from the preamble. Contention and transmission must therefore fit within a bounded window of

$$W_{\text{eff}} \;=\; \left\lfloor \frac{D_{\text{rem}} - 2\,\delta_{\text{sw}}}{\sigma} \right\rfloor \quad \text{slots}, \tag{1}$$

where σ is the slot time, typically 9 μs in IEEE 802.11-compliant systems. Hence, a single NPCA transition admits NPCA transmissions during W_{eff} slots at most, which we index by $t \in \{1, 2, \ldots, W_{\text{eff}}\}$.

This deadline is the crux of NPCA. Every backoff decision competes against it. A STA that collides doubles its contention window as usual, and once the resulting backoff exceeds the slots left in W_{eff}, it can no longer transmit before the window closes so it wastes the rest of the opportunity despite having data to send.

Because the contention window is reset at every NPCA visit, this penalty recurs each time rather than amortizing across visits. This mismatch between open-ended backoff and the finite window is the inefficiency that PACE targets.

C. Slotted Contention Model

We model time on the NPCA primary channel as a sequence of slots of duration σ. Each visitor maintains a per-slot transmission probability $\tau_i(t)$ at each time slot t and, when eligible, transmits in slot t with that probability, in the manner of slotted ALOHA; we abbreviate it as τ_i where the slot index is immaterial. This per-slot access probability is the variable that the scheme of Section IV controls. The standard baseline instead runs its backoff countdown unchanged, and enters the model only through the slot outcomes defined below.

This probabilistic access departs from the current draft, which reuses the enhanced distributed channel access (EDCA) backoff procedure on the NPCA primary channel [1]. PACE keeps the NPCA architecture intact, including the switching triggers, the NPCA timer, and the contention-state save and restore, and replaces only the contention rule applied within the bounded visit.

Following the slot indexing $t \in \{1, \ldots, W_{\text{eff}}\}$ of Section III-B, we denote by $W_{\text{rem}}(t) = W_{\text{eff}} - t + 1$ the number of slots remaining at slot t, including the current one, so that W_{rem} counts down toward the deadline as contention proceeds. All STAs are assumed to be saturated, so each STA i always holds a pending head-of-line PPDU, whose transmission occupies L_i slots, and a STA that completes a transmission draws its next frame and continues contending. L_i therefore changes only at STA i's own delivery instants and is constant in between; we suppress this time dependence in the notation and let L_i always denote the current head-of-line frame. Because a transmission must complete before the window closes, STA i is viable at slot t only if its frame still fits, i.e. $L_i \le W_{\text{rem}}(t)$; otherwise it can no longer succeed within the visit. We write the viable population at slot t as

$$\mathcal{V}(t) \;=\; \big\{\, i \in \mathcal{V} : L_i \le W_{\text{rem}}(t) \,\big\}, \tag{2}$$

which is non-increasing in t as both the window depletes and frames become infeasible.

At each slot t, a viable STA i attempts transmission with probability $\tau_i(t)$. Following the half-duplex assumption, a

[1]We use the term BSS primary channel to distinguish the original primary channel from the NPCA primary channel introduced in IEEE 802.11bn. Prior to 802.11bn, when NPCA was not yet defined, this channel was referred to simply as the primary channel.

[2]NPCA relies on preamble detection rather than energy detection. A STA must decode the inter-BSS PPDU preamble at PHY-RXSTART to both classify the transmission and derive its remaining duration, from which the NPCA window is determined. Energy detection alone, which yields no length or BSS information, is insufficient to trigger an NPCA transition.

transmitting STA cannot sense the channel in the same slot. Let $\mathcal{D}(t)$ denote the set of STAs whose transmissions on the NPCA primary channel are detected by a given STA in slot t. This set is defined at the channel level. It counts every STA transmitting on that channel in the slot, not only the switching visitors but also any other STA that uses the same channel as its own primary. The slot outcome is idle when $|\mathcal{D}(t)| = 0$, a success when $|\mathcal{D}(t)| = 1$, and a collision when $|\mathcal{D}(t)| \geq 2$. These three observable events are the only feedback available to a STA, and they drive the adaptive contention control developed in Section IV.

The remaining window advances by the time each outcome occupies the medium. An idle slot consumes one slot, while a successful slot consumes L_{solo} slots and a collision consumes L_{col} slots. The two occupancies depend on the access mode.

1) Basic Access: A successful slot is occupied by the sole sender's frame,

$$L_{\text{solo}} \;=\; L_i \quad \text{if } \mathcal{D}(t) = \{i\}, \tag{3}$$

while a collision keeps the medium busy until the longest colliding frame has ended,

$$L_{\text{col}} \;=\; \max_{i \in \mathcal{D}(t)} L_i, \tag{4}$$

as in standard IEEE 802.11 without collision detection. Since each STA contends for one head-of-line PPDU at a time, L_i is fixed until that frame is delivered; the slot dependence of both occupancies enters through the identity of the transmitters $\mathcal{D}(t)$ and their current frames.

2) RTS/CTS-Protected Access: Under RTS/CTS-protected access the busy-slot accounting changes. A successful exchange is preceded by the handshake and therefore occupies

$$L_{\text{solo}} \;=\; L_i + L_{\text{hs}} \quad \text{if } \mathcal{D}(t) = \{i\}, \tag{5}$$

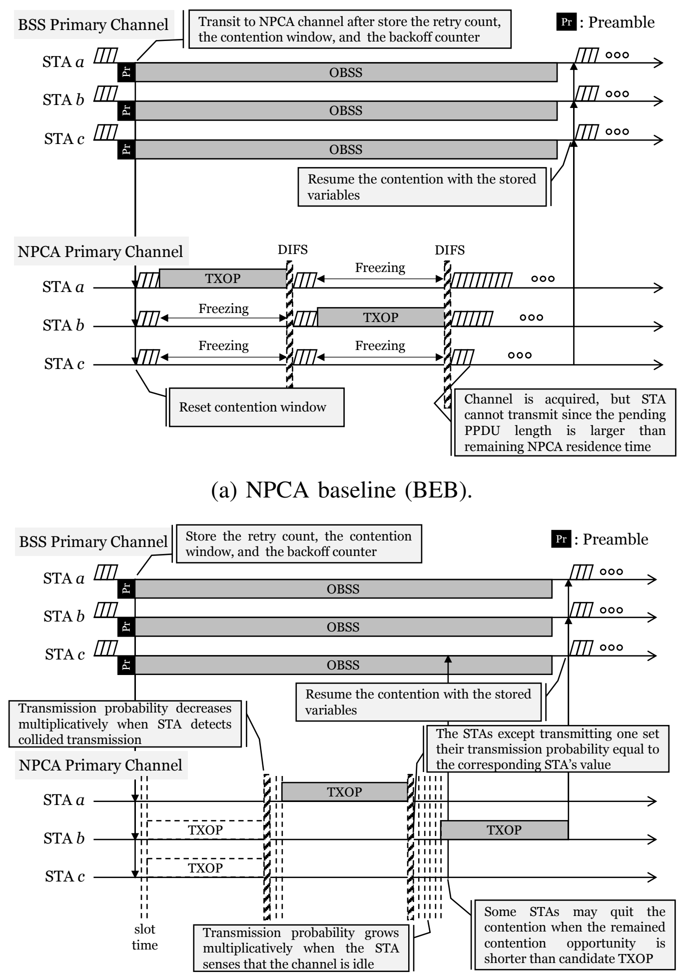
slots, where

$$L_{\text{hs}} \;=\; \left\lceil \frac{T_{\text{RTS}} + T_{\text{CTS}} + 2\,T_{\text{SIFS}}}{\sigma} \right\rceil \tag{6}$$

is the handshake overhead, with T_{RTS}, T_{CTS}, and T_{SIFS} being the RTS, CTS, and short interframe space (SIFS) durations defined in IEEE 802.11. A collision, by contrast, involves only RTS frames. The data frames are never sent, and the channel recovers after the extended interframe space, so a collision occupies

$$L_{\text{col}} \;=\; \left\lceil \frac{T_{\text{RTS}} + T_{\text{EIFS}}}{\sigma} \right\rceil \tag{7}$$

slots, where $T_{\text{EIFS}} = T_{\text{SIFS}} + T_{\text{ACK}} + T_{\text{DIFS}}$ is the extended interframe space with T_{ACK} and T_{DIFS} being the acknowledgment (ACK) and distributed interframe space (DIFS) durations defined in IEEE 802.11. Unlike its basic-access counterpart of (4), this occupancy is a constant, independent of the colliding frame lengths and of the slot index. Under protection the viability test of (2) must also leave room for the handshake, i.e. L_i is replaced by $L_i + L_{\text{hs}}$. Basic access is recovered as the special case $L_{\text{hs}} = 0$ with L_{col} given by (4). Both access modes are evaluated in Section V-A.



(a) NPCA baseline (BEB).

(b) Proposed PACE (probabilistic adaptive contention).

Fig. 3: Medium access on the NPCA primary channel: a STA that detects an OBSS transmission on the BSS primary channel migrates and contends within the bounded NPCA window. (a) the standard NPCA baseline with binary exponential backoff, and (b) the proposed PACE with probabilistic adaptive contention and PPDU-aware self-exclusion.

IV. PACE Algorithm

Fig. 3 illustrates contention on the NPCA primary channel under the slotted model of Section III-C and contrasts the two schemes studied in this paper. In both, a STA that detects an OBSS transmission on the BSS primary channel stores its DCF state, migrates to the NPCA primary channel, and contends there until the NPCA timer expires, whereupon it restores the saved state and resumes on the BSS primary channel. Under the NPCA baseline of Fig. 3a, STAs contend with BEB. A STA that collides doubles its contention window, so its counter can outlast the remaining window, and because the countdown is frozen during every busy period, the surviving counters tend to expire together and collide again.

Meanwhile, under the proposed PACE scheme of Fig. 3b, each STA instead adapts its per-slot probability τ_i from the idle, success, and collision feedback. The access probability is raised multiplicatively on idle slots, lowered on collisions, and

Algorithm 1 PACE at STA i during one NPCA visit

Require: head-of-line PPDU length L_i (slots), window W_{eff}, factors c_{coll}, $c_{\mathrm{idle}} > 1$, handshake overhead L_{hs} of Eq. (6) ($L_{\mathrm{hs}} = 0$ under basic access)

/* Phase (a): Probability Initialization */

$\tau_i \leftarrow 1/W_{\mathrm{eff}}$; $\quad W_{\mathrm{rem}} \leftarrow W_{\mathrm{eff}}$

while $W_{\mathrm{rem}} > 0$ **do**

  /* Phase (b): PPDU-aware Self-Exclusion */

  if $L_i + L_{\mathrm{hs}} > W_{\mathrm{rem}}$ **then**

    $\tau_i \leftarrow 0$

  end if

  /* Phase (c): Probabilistic Channel Access */

  draw $u \sim \mathrm{Uniform}(0, 1)$

  if $u < \tau_i$ **then**

    transmit the PPDU, advertising τ_i (preceded by the RTS/CTS handshake under protected access)

    if solo success **then**

      $W_{\mathrm{rem}} \leftarrow W_{\mathrm{rem}} - L_{\mathrm{solo}}$; $\quad L_i \leftarrow$ next pending PPDU

    else

      $W_{\mathrm{rem}} \leftarrow W_{\mathrm{rem}} - L_{\mathrm{col}}$ {collided: Eq. (7) under protection, Eq. (4) under basic access}

    end if

  else

    /* Phase (d): Feedback-based Adaptation */

    observe the slot outcome $|\mathcal{D}(t)|$

    if $|\mathcal{D}(t)| = 0$ **then**

      $\tau_i \leftarrow \min(\tau_i \cdot c_{\mathrm{idle}},\ 1)$; $\quad W_{\mathrm{rem}} \leftarrow W_{\mathrm{rem}} - 1$

    else if $|\mathcal{D}(t)| = 1$ where $\mathcal{D}(t) = \{j\}$ **then**

      $\tau_i \leftarrow \tau_j$ {the solo sender's advertised probability}; $W_{\mathrm{rem}} \leftarrow W_{\mathrm{rem}} - L_{\mathrm{solo}}$

    else

      $\tau_i \leftarrow \tau_i / c_{\mathrm{coll}}$; $\quad W_{\mathrm{rem}} \leftarrow W_{\mathrm{rem}} - L_{\mathrm{col}}$

    end if

  end if

end while

replaced by the successful sender's value on solo receptions. In addition, a STA proactively withdraws from contention once its frame can no longer fit within $W_{\mathrm{rem}}(t)$.

Algorithm 1 summarizes PACE as executed independently by each STA on the NPCA primary channel during a single NPCA visit. Every STA runs the same rule using only locally observable feedback, those are, the per-slot outcome and its own remaining window without central coordination or knowledge of the contending population.

We adopt the probabilistic approach for two reasons. First, the per-slot probability τ is the common abstraction under which DCF, PND, and AND are all analyzed [5], [9], [15], so it places PACE on the same analytical footing as these schemes. Second, and more fundamentally, the finite NPCA window is precisely where a deterministic countdown breaks down. A single backoff draw cannot re-adapt within the handful of slots a visit lasts, whereas a per-slot probability can be re-steered every slot by the feedback law, which will be described in Subsec. IV-D. The ALOHA-style attempt is therefore not incidental but the mechanism that makes within-window adaptation possible.

Algorithm 1 runs four phases per visit. (a) a one-time probability initialization, followed by a per-slot loop of (b) self-exclusion, (c) channel access, and (d) feedback-based adaptation. The remainder of this section develops each phase in turn.

A. Phase (a): Probability Initialization

A visitor that has just switched to the NPCA primary channel does not know how many other STAs are contending there. The population-matched choice $\tau_0 = 1/(N_{\mathrm{vis}}+N_{\mathrm{nat}})$, which is the well-known throughput-optimal attempt probability of slotted ALOHA, is therefore unavailable in practice since the contender count is not observable.

Visitor STAs might, through association state or management traffic, form some estimate of how many STAs belong to its own BSS. The harder obstacle is that the population it actually contends with also includes the native STAs of the neighboring BSS that owns the NPCA primary channel. That is, a visitor competes against these as well, yet has no direct way to learn how many are active. The count that would parameterize $1/(N_{\mathrm{vis}}+N_{\mathrm{nat}})$ is thus doubly hidden, which motivates a surrogate that avoids counting STAs altogether.

We propose a method where the window length itself determines the scale. A visit spans W_{eff} slots, and a STA contends in at most one of them per slot so W_{eff} upper-bounds the number of attempt opportunities any STA sees in a visit. A STA attempting with probability τ therefore expects at most $W_{\mathrm{eff}}\,\tau$ attempts before the deadline. Fixing this budget to a single probe per STA, $W_{\mathrm{eff}}\,\tau_0 = 1$ sets the most conservative self-consistent exploration rate. Low enough to avoid an early collision storm, yet high enough for each STA to test the medium about once before the window drains. This yields

$$\tau_0 = \frac{1}{W_{\mathrm{eff}}}, \tag{8}$$

with which each visitor starts the visit, $\tau_i(1) = \tau_0$. It depends only on the effective window W_{eff}, which is a quantity that each STA can compute locally from the OBSS PPDU duration and no knowledge of the contender count is required.

In general frame length to be transmitted on NPCA primary channel is longer than a single slot, so τ_0 is bound to be smaller than the optimal value. This choice is deliberately biased toward under-attempting for two reasons. First, it is self-correcting in the safe direction. An over-conservative τ_i is raised toward the operating point within a few slots by the idle-driven multiplicative increase in Phase (d), whereas an over-aggressive start triggers collisions whose wasted slots cannot be reclaimed inside a finite window. The MIMD recovery is thus asymmetric, and starting low pays only a bounded ramp-up cost. Second, $1/W_{\mathrm{eff}}$ tracks the regime where initialization matters most. When the window is tight, i.e., $W_{\mathrm{eff}} \approx N_{\mathrm{vis}}+N_{\mathrm{nat}}$, it already approximates the optimum $1/(N_{\mathrm{vis}}+N_{\mathrm{nat}})$, and when the window is loose such as $W_{\mathrm{eff}} \gg N_{\mathrm{vis}}+N_{\mathrm{nat}}$, the mild initial timidity is erased by the idle increase long before the window drains.

B. Phase (b): PPDU-aware Self-Exclusion

At the start of every slot a STA applies a frame-fit test, comparing the airtime its own frame needs to the remaining window $W_{\mathrm{rem}}(t)$. Following the viability condition of Eq. (2), that airtime is L_i under basic access and $L_i + L_{\mathrm{hs}}$ under RTS/CTS protection, since the handshake of Eq. (6) must also fit before the deadline. If the frame can no longer finish, the STA sets $\tau_i(t) = 0$ and stays out of the slot. The test is repeated at every subsequent slot, and because $W_{\mathrm{rem}}(t)$ only shrinks the STA remains silent for the rest of the visit unless it later holds a shorter frame. The test involves only the STA's own frame and the slot index and thus, like initialization, requires no knowledge of the contending population.

Conventional NPCA has no such rule, yet the deadline binds it all the same, since a STA shall switch back to the BSS primary channel when the NPCA timer expires [1]. A frame that does not fit the remaining window is therefore cut off at expiry even if it wins the channel; self-exclusion makes this inevitability explicit and returns those slots to the STAs that can still finish.

Self-exclusion keeps the optimal target that PACE aims to track,

$$\tau^*(t) \;=\; \frac{1}{|\mathcal{V}(t)|}, \tag{9}$$

meaningful in a heterogeneous and finite window. In a slot with $|\mathcal{V}(t)|$ viable STAs, a success, meaning exactly one transmitter, is most likely when each attempts with probability $\tau^*(t)$, and this target rises over the visit as the viable set drains. A STA whose frame no longer fits is removed from the viable set $\mathcal{V}(t)$ of Eq. (2), so it neither wastes time on attempts destined to fail nor inflates the contention seen by the STAs that can still succeed. This plays the same role as the collision detection departure of a previously proposed neighbor discovery algorithm [5], in which a STA leaves the contention once its advertisement succeeds. Here a STA also leaves once its frame provably cannot complete, tightening $\mathcal{V}(t)$ toward the set of STAs that genuinely compete for the remaining slots.

C. Phase (c): Probabilistic Channel Access

A STA that is still viable transmits in slot t with probability $\tau_i(t)$ and otherwise listens. Because the radio is half-duplex, a transmitting STA cannot sense the slot it occupies. It learns from the acknowledgement whether its own frame got through, which tells it how far the window has advanced and whether to dequeue the next frame, but it observes none of the three slot outcomes that drive the adaptation, so it leaves τ_i untouched on its own transmissions. Each transmitted PPDU advertises the sender's current $\tau_i(t)$ in its header[3], which the listeners use in Phase (d). This value describes only the sender's own state and rides in a frame it transmits anyway, in the manner of the per-frame information IEEE 802.11 headers already carry, so no control frame is added and no STA ever learns how many others are contending.

A slot is a success when exactly one viable STA transmits and its frame fits the remaining window. Being saturated, the winner then draws its next pending frame and re-enters the contention with its current τ_i, which the solo advertisement has just propagated to the listeners; the viable set $\mathcal{V}(t)$ therefore shrinks only through self-exclusion as the deadline nears. Slots with no transmission are idle, and slots with two or more transmissions are collisions; both outcomes are observed by every non-transmitting STA and feed the adaptation of Phase (d).

D. Phase (d): Feedback-based Adaptation

Every STA that listened in slot t updates its probability from the slot outcome $|\mathcal{D}(t)|$ using the MIMD rule inherited from neighbor discovery as

$$\tau_i(t') = \begin{cases} \min\bigl(c_{\mathrm{idle}}\, \tau_i(t),\, 1\bigr) & |\mathcal{D}(t)| = 0 \text{ (idle)}, \\ \tau_j(t) & \mathcal{D}(t) = \{j\} \text{ (success)}, \\ \tau_i(t)/c_{\mathrm{coll}} & |\mathcal{D}(t)| \geq 2 \text{ (collision)}, \end{cases} \tag{10}$$

with $c_{\mathrm{idle}}, c_{\mathrm{coll}} > 1$, where t' denotes the next contention slot after the outcome of slot t, i.e., $t' = t + 1$ after an idle slot and $t' = t + L_{\mathrm{solo}}$ or $t' = t + L_{\mathrm{col}}$ after a busy one, and τ_i is unchanged while the medium is occupied.

An idle slot signals that the population has been overestimated and the attempt rate is raised by the factor of c_{idle}. On the other hand, a collision signals the opposite, hence, once STAs detect a collision, the attempt rate is divided by c_{coll} to reduce the contention. On a solo success, every listener copies the advertised τ_j, so a single successful slot propagates the winning operating point to the whole contending set at once, giving PACE the fast intra-visit consensus.

None of the three updates counts the viable STAs. Instead, the idle and collision statistics move each τ_i toward the operating point implicitly, and solo-copy pulls the population onto a common value, so the ensemble approximately tracks the time-varying optimum $\tau^*(t)$ of Eq. (9) without estimating the number of contending STAs $|\mathcal{V}(t)|$.

The direction of this adaptation is what separates PACE from BEB. Because the viable set only shrinks over a visit, the optimal target $\tau^*(t) = 1/|\mathcal{V}(t)|$ of (9) rises as the deadline approaches. On the other hand, the BEB moves the opposite way, that is, a collision doubles the contention window and halves the attempt rate, so the access probability falls exactly when it should climb.

Finally, the rule of Eq. (10) needs no modification to operate alongside the native STAs of Section III-A. Because the slot outcome $\mathcal{D}(t)$ is measured at the channel level, native activity enters a visitor's feedback directly. A slot in which two or more frames overlap is a collision whether the extra transmitter is a visitor or a native, so a native transmission that clashes with a visitor lowers that visitor's τ_i, and a slot with no transmission at all, native or visitor, is idle and raises it. The solo case is the one exception. The copy $\tau_i(t') = \tau_j(t)$ requires the successful frame to advertise a probability, which only a

[3] A few quantization bits for τ_i fit in the Control Information field of a newly defined A-Control subfield, identified by a new Control ID, of the HT Control field that rides unencrypted in the MAC header of a QoS Data frame. Defining such a subfield would require an IEEE 802.11 amendment, but follows the existing A-Control extension framework. One placement serves both access modes, since RTS and CTS carry no HT Control field and a solo success propagates τ_i only once the QoS Data MPDU is decoded.

PACE visitor does; when a native transmits alone, its frame carries no such value, so a listening visitor keeps τ_i unchanged and simply observes the busy slot while the window continues to shrink.

E. Window Scaling of the Adaptation Factors

The rule of Eq. (10) leaves the factors c_{idle} and c_{coll} open, and the finite window turns their choice into a trade-off between two losses that both accumulate before the deadline. A small factor adapts slowly and spends the visit ramping toward the operating point, whereas a large one reaches it quickly but keeps overshooting around it. This subsection quantifies the balance by treating Phase (d) as a finite-horizon stochastic approximation. We set the two factors equal, which keeps every τ_i on a common multiplicative grid so that the solo-copy of Eq. (10) exchanges values the listeners can themselves reach, and the pair then reduces to a single log-domain step size

$$\epsilon \;=\; \ln c_{\text{idle}} \;=\; \ln c_{\text{coll}}. \tag{11}$$

Let $X_k = \ln \tau_i$ at the k-th update opportunity of a visit. Under Eq. (10) the state performs a random walk $X_{k+1} = X_k + \epsilon H_k$, where H_k equals $+1$ on an idle outcome, -1 on a collision, and 0 otherwise. The mean drift $h(x) = p_{\text{I}}(x) - p_{\text{C}}(x)$, the gap between the idle and collision probabilities seen from state x, vanishes at the operating point x^* that the feedback seeks. Near a stable x^* the error $Y_k = X_k - x^*$ follows the linearized recursion

$$Y_{k+1} \;\approx\; (1-\kappa\epsilon)\,Y_k + \epsilon\,\xi_{k+1}, \tag{12}$$

with restoring rate $\kappa = -h'(x^*) > 0$ and zero-mean outcome noise ξ of variance $\nu^2 = p_{\text{I}}(x^*) + p_{\text{C}}(x^*)$.

A visit is scored by what it accumulates rather than by where it ends. We make the objective explicit as

$$J \;=\; \ln T \;-\; \alpha\left(\ln\rho\right)^2, \tag{13}$$

where T is the fraction of the window carrying useful airtime, ρ is the visitors' share of that airtime divided by their population share $N_{\text{vis}}/(N_{\text{vis}}{+}N_{\text{nat}})$, and $\alpha \geq 0$ weights proportionality against efficiency. The quadratic log penalty vanishes exactly at the proportional share $\rho = 1$ and charges over- and under-service symmetrically, so J scores the design goal of Section IV-B, an efficient window that neither starves the visitors nor lets them displace the natives. Both ingredients accumulate over the whole window and are the measures that Section V reports. Treating J as a function of the operating point and assuming it is locally quadratic around x^* with curvature $q > 0$, summing the second moment of Eq. (12) over the K update opportunities of a visit gives the loss

$$R_K(\epsilon) \;\approx\; \frac{q}{4\kappa}\left(\frac{Y_0^2}{\epsilon} + K\nu^2\epsilon\right), \tag{14}$$

where Y_0 is the initial offset. The first term is the transient cost of a slow ramp and decays as $1/\epsilon$, while the second is the fluctuation that a large step sustains around the operating point and grows as $K\epsilon$. The two scale oppositely in ϵ, and balancing them yields

$$\epsilon^* \;=\; \frac{|Y_0|}{\nu\sqrt{K}}. \tag{15}$$

The number of update opportunities is in turn set by the window. Every outcome occupies the medium for one idle slot, L_{solo} slots, or L_{col} slots, so $K \approx W_{\text{eff}}/\bar{L}$, where $\bar{L}$ is the mean outcome occupancy of the access mode in use. Substituting into Eq. (15),

$$\epsilon^* \;=\; \frac{C}{\sqrt{W_{\text{eff}}}}, \qquad C \;=\; \frac{|Y_0|\sqrt{\bar{L}}}{\nu}, \tag{16}$$

which prescribes $c^*_{\text{idle}} = c^*_{\text{coll}} = \exp\bigl(C/\sqrt{W_{\text{eff}}}\,\bigr)$. The constant also separates the two access modes. Since basic access and RTS/CTS protection differ through L_{solo} and L_{col}, their constants relate as $C_{\text{RTS}}/C_{\text{basic}} \approx \sqrt{\bar{L}_{\text{RTS}}/\bar{L}_{\text{basic}}}$, so the mode whose outcomes hold the channel longer, and hence grants fewer updates per window, calls for a proportionally larger step. The argument is summarized as follows.

Proposition 1: Consider the finite-horizon adaptation $X_{k+1} = X_k + \epsilon H_k$ with a locally stable equilibrium x^*, a locally quadratic utility, and bounded outcome noise of variance ν^2. If a visit contains K effective update opportunities and both the transient and the fluctuation loss of Eq. (14) are accumulated, the loss-minimizing constant step satisfies

$$\epsilon^*_K \;=\; \frac{|X_0 - x^*|}{\nu\sqrt{K}} + o\bigl(K^{-1/2}\bigr),$$

hence $\epsilon^*_K = \Theta(K^{-1/2})$. If moreover $K = W_{\text{eff}}/\bar{L} + o(W_{\text{eff}})$, then $\epsilon^* = C/\sqrt{W_{\text{eff}}} + o\bigl(W_{\text{eff}}^{-1/2}\bigr)$ with C as in Eq. (16).

Two remarks bound the claim. First, Phase (a) sets $X_0 = -\ln W_{\text{eff}}$, so the offset $|Y_0| = |\ln(W_{\text{eff}}\,\tau^*)|$ retains a slow logarithmic dependence on the window, and C behaves as a constant only over ranges of W_{eff} in which this variation is small enough to be absorbed. Eq. (16) is therefore an asymptotic inverse-square-root scaling under the local finite-horizon approximation, not an exact equality. Second, the scaling presumes the cumulative criterion above; a criterion that graded only the terminal state would favor a smaller step of order $\ln K/K$. Within these bounds, Eq. (16) delivers the same message as the initialization of Phase (a). Both design knobs of PACE are set from the window alone, the starting probability at $1/W_{\text{eff}}$ and the adaptation step at $\Theta(1/\sqrt{W_{\text{eff}}})$, with faster adaptation in shorter visits.

The scaling is moreover directly implementable, because W_{eff} is not merely observable but signaled. The OBSS frame that triggers the switch announces its remaining duration in its preamble, from which every visitor already computes W_{eff} through Eq. (1), so a factor of the form $\exp\bigl(C/\sqrt{W_{\text{eff}}}\,\bigr)$ is a runtime value obtained at the start of each visit rather than a constant fixed at deployment. We call this variant *PACE-dynamic*, with the single constant C calibrated once at a reference window to maximize the objective of Eq. (13) at $\alpha = 0.5$, and refer to the fixed-factor configuration as *PACE-static*. The names describe only how the step size is chosen. Both variants adapt τ_i on every observed slot, and both draw on the same local information as the initialization of Phase (a).

TABLE II: Simulation parameters. Visitors run the scheme under test, natives always run standard DCF, and the RTS/CTS costs assume a 24 Mbps control rate.

Parameter	Value
Visitor STAs N_{vis}	$\{5, 10, 20, 50\}$
Native STAs N_{nat}	10
Slot time σ / SIFS / DIFS	9 / 16 / 34 μs
Effective window W_{eff}	420 slots (3.78 ms)
Visitor PPDU duration	$\mathcal{U}[225, 900]$ μs
Native PPDU duration	450 μs
Initial probability τ_0	$1/W_{\mathrm{eff}}$
MIMD factors $c_{\mathrm{coll}}, c_{\mathrm{idle}}$ (PACE-static)	1.5
Step-size constant C (PACE-dynamic)	10.16
$\mathrm{CW}_{\min}$ / $\mathrm{CW}_{\max}$ for BEB	15 / 1023
RTS/CTS L_{hs} / L_{col}	88 / 106 μs
Transitions / sequences / seeds	50 / 20 / 5

Section V-C compares the two across the window range, where a single fixed factor turns out to be too small for a short visit and too large for a long one.

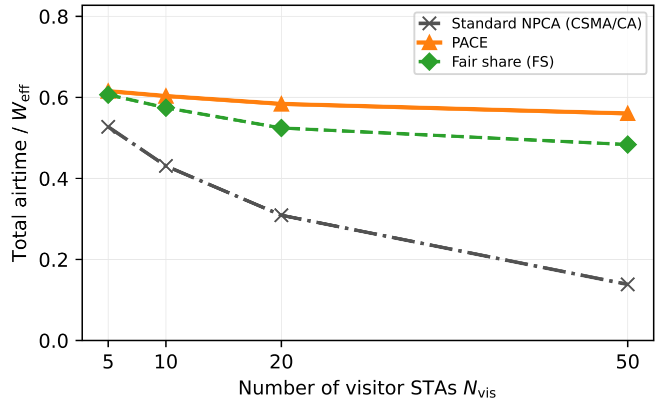
V. Performance Evaluation

We evaluate PACE in the slotted NPCA contention model under IEEE 802.11 orthogonal frequency division multiplexing (OFDM) timing. Table II summarizes the parameters. A visit spans $W_{\mathrm{eff}} = 420$ slots of $\sigma = 9$ μs, which corresponds to 3.78 ms, and is contended by $N_{\mathrm{vis}} \in \{5, 10, 20, 50\}$ visitor STAs running the scheme under test alongside $N_{\mathrm{nat}} = 10$ native STAs that always run standard DCF. Visitor PPDU durations are drawn uniformly from 225–900 μs and native frames last 450 μs. PACE follows Algorithm 1 with $\tau_0 = 1/W_{\mathrm{eff}}$. Unless stated otherwise, PACE denotes PACE-static with the fixed factors $c_{\mathrm{coll}} = c_{\mathrm{idle}} = 1.5$; PACE-dynamic, which recomputes the factors per visit as $c = \exp\bigl(C/\sqrt{W_{\mathrm{eff}}}\,\bigr)$ with $C = 10.16$ calibrated at the reference window, where it yields $c = 1.64$, enters the comparison in Section V-C, in which the window varies.

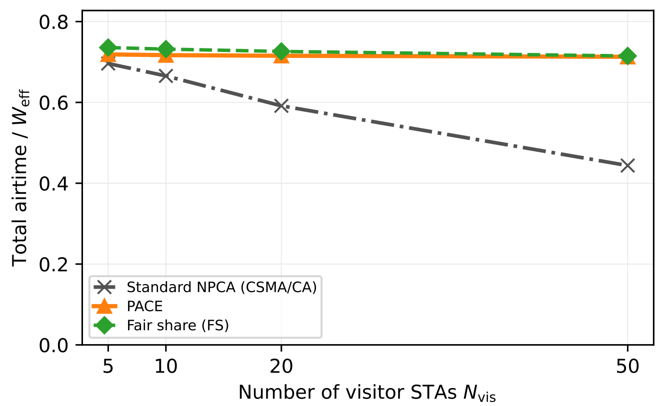
We compare PACE against two references. The first is the standard-compliant NPCA baseline, which contends with carrier sense multiple access with collision avoidance (CSMA/CA) under $\mathrm{CW}_{\min} = 15$ and binary exponential back-off (BEB) as Draft 1.2 specifies [1]. Since the draft gates only the switching decision through the NPCA Minimum Duration Threshold and leaves the unfittable-frame case unspecified, we grant the baseline the stronger conformant behavior, a deferring implementation that withholds frames exceeding the remaining window through the same locally computable test as PACE's Phase (b).

The second is a genie-aided *fair-share* reference (FS) that transmits at $\tau^*(t) = 1/|\mathcal{V}(t)|$ with perfect knowledge of the viable set. FS is not an airtime optimum but the attempt-fair operating point that PACE is designed to track.

Native STAs, which neither know nor observe the visitors' window, keep their standard DCF parameters unchanged in every scheme and contend without any fit restriction; a native frame that straddles the window end completes on the channel, with its in-window portion credited to the window utilization. Both access modes of Section III-C are evaluated, basic access charging a collision the longest colliding frame of Eq. (4), and mandatory RTS/CTS with the standard-derived costs $L_{\mathrm{hs}} = 88$ μs and $L_{\mathrm{col}} = 106$ μs. Each data point averages the steady half of 50 consecutive NPCA transitions over 20 sequences and five seeds. Performance is reported as the total useful airtime of the channel normalized by W_{eff}, its split between visitors and natives, and the visitor airtime proportionality index defined in Section V-B; the two axes are those of the objective J of Eq. (13).



(a) Basic access (no RTS/CTS).



(b) Mandatory RTS/CTS.

Fig. 4: Total useful airtime (visitor and native combined) versus the number of visitor STAs N_{vis} for standard-compliant NPCA (CSMA/CA visitors), PACE, and the fair-share reference (FS), in the mixed native/visitor channel with $N_{\mathrm{nat}} = 10$ native DCF stations.

A. Why PACE: Channel Airtime under Mixed Contention

We first ask the question that motivates this work. What does PACE buy over simply running the standard access procedure on the non-primary channel? The design objective of PACE is not visitor throughput per se, but to keep the shared NPCA channel efficient and fair. The FS policy $\tau^*(t) = 1/|\mathcal{V}(t)|$, computed over all contenders, is precisely the embodiment of that objective, and the evaluation therefore reports the total useful airtime of the channel.

Fig. 4 compares $N_{\mathrm{vis}} \in \{5, \ldots, 50\}$ visitor STAs operating either standard CSMA/CA or PACE exactly as specified in

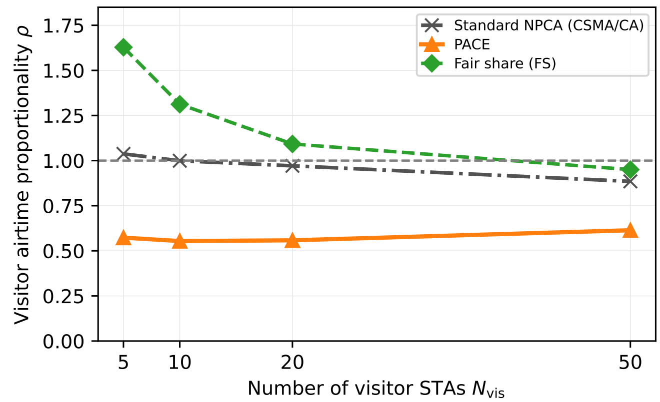
Algorithm 1, every visit starts cold from $\tau_0 = 1/W_{\text{eff}}$, sharing the channel with $N_{\text{nat}} = 10$ native DCF stations, using IEEE 802.11 OFDM timing. Under basic access in Fig. 4a, the channel under standard-compliant visitors degrades monotonically as the visitor population grows. When tens of STAs draw their backoff from the same fixed $\text{CW}_{\min} = 16$, several counters reach zero in the same slot, and the resulting collision occupies the channel for the full duration of the longest colliding frame. Over an unbounded horizon this would be a transient, since BEB keeps doubling the windows until the attempt rate settles near a level the population can sustain. An NPCA visit, however, ends within a few milliseconds, and the adjustment BEB needs grows with the contending population while the time available for it stays fixed. The larger N_{vis} becomes, the greater the fraction of the window that is spent on collisions before BEB has adjusted, so the standard procedure degrades not only the visitors' own throughput but the total useful airtime of the shared channel, which falls to 0.14 at $N_{\text{vis}} = 50$.

PACE instead sustains a total airtime of 0.56–0.62 across the entire range, above FS at every point, and reaches $4.1\times$ the standard procedure at $N_{\text{vis}} = 50$. This stability reflects the intended fair-sharing behavior. At $N_{\text{vis}} = 5$ the conservative initialization $\tau_0 = 1/W_{\text{eff}}$ leaves most of the channel to the natives, and as the visitor population and its offered load grow, the idle-driven increase raises the visitors' share monotonically, while self-exclusion releases slots that can no longer carry a full frame. PACE exceeds FS in total airtime under this access mode because FS enforces the per-STA fair attempt rate at the cost of collisions that basic access charges at full frame length, whereas PACE's gradual probability increase avoids them.

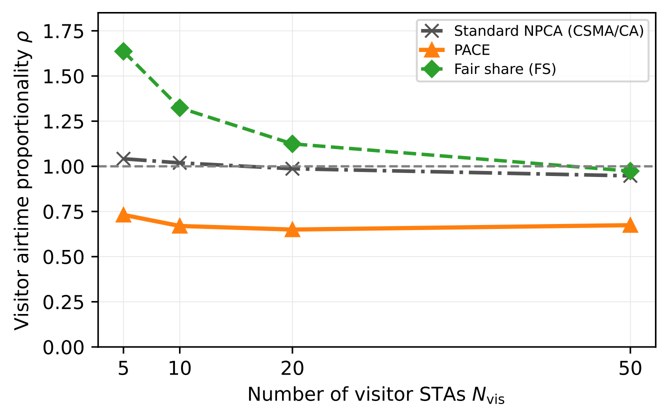
Under RTS/CTS-protected access, shown in Fig. 4b, a collision costs the constant L_{col} of Eq. (7), which is 106 μs, instead of a full frame, at the price of the handshake overhead L_{hs} of 88 μs on every success. Since a collision now occupies the channel only briefly, the fair-share policy is also airtime-efficient, and PACE tracks it within 3% over the whole range, holding 0.71–0.72, while the channel under standard visitors declines from 0.70 to 0.44. The handshake bounds the duration of a collision, but it mitigates neither the frozen-backoff idling nor the synchronized collisions, so PACE retains 61% more channel airtime at $N_{\text{vis}} = 50$. Inside a finite NPCA window shared with natives, probabilistic deadline-aware access keeps the channel near the efficient and fair operating point, whereas the standard backoff loses the same airtime to idle slots and collisions.

B. Fairness between Natives and Visitors

Total airtime alone does not reveal who receives it, so we next examine how the window is divided between the two populations. Fig. 5 reports the visitor airtime proportionality index ρ, the visitors' share of the useful airtime normalized by their population share $N_{\text{vis}}/(N_{\text{vis}} + N_{\text{nat}})$, in the identical setting as Fig. 4. $\rho = 1$ is proportional fairness, and $\rho > 1$ indicates that the visitors occupy the channel beyond their share.



(a) Basic access (no RTS/CTS).



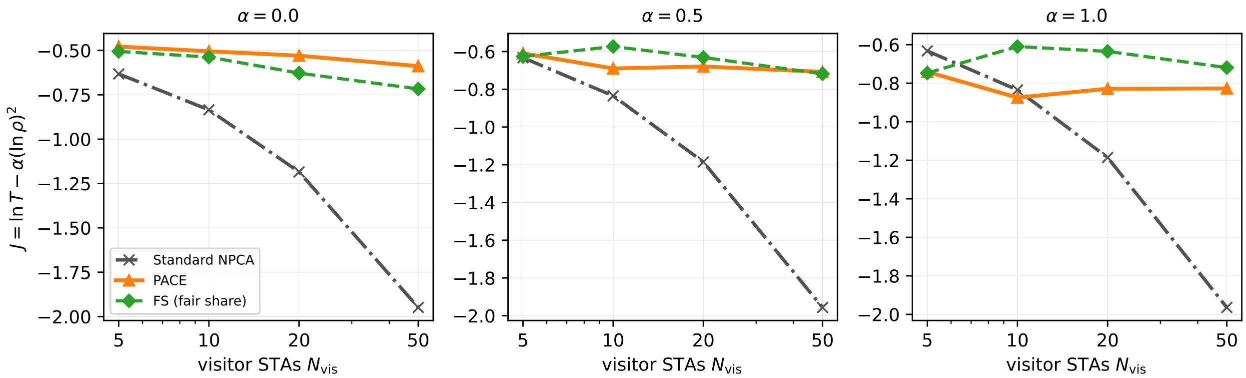
(b) Mandatory RTS/CTS.

Fig. 5: Visitor airtime proportionality $\rho = (\text{visitor airtime share})/(N_{\text{vis}}/(N_{\text{vis}} + N_{\text{nat}}))$ versus the number of visitor STAs N_{vis}, in the same setting as Fig. 4. $\rho = 1$ is the proportional-fair reference; $\rho > 1$ means the visitors occupy more than their population share.

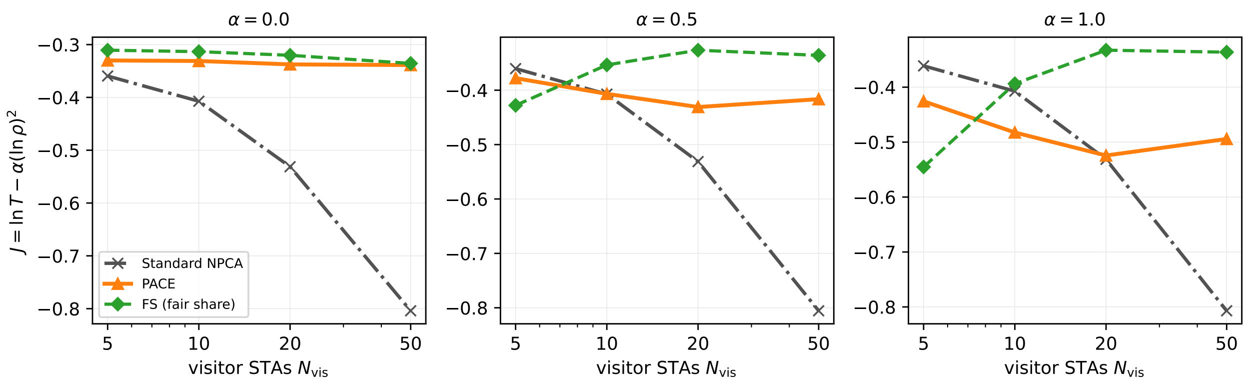
FS itself starts at $\rho = 1.6$ for $N_{\text{vis}} = 5$ and falls to about 0.95 at $N_{\text{vis}} = 50$. Even a per-STA fair attempt rate grants the visitors more than their population share because the natives' DCF decrements its backoff only in idle slots and therefore attempts less frequently inside a busy window than a per-slot probability does. Exact proportionality is thus unattainable by any attempt-fair policy at low demand and the FS curve marks the practically fair operating line.

Against this reference the two schemes separate clearly. The standard-compliant visitors hold $\rho \approx 0.9$–1.0, but this is a proportional share of a channel whose total airtime their collisions have already reduced at large N_{vis}. PACE operates below unity throughout, holding ρ at 0.55–0.61 under basic access and 0.65–0.73 under RTS/CTS, so the visitors settle at roughly two thirds of their proportional share regardless of their population.

The visitors are guests on the primary channel of the neighboring BSS, so an NPCA scheme should let them exploit idle capacity without displacing the native traffic. It attains the highest total airtime among the three schemes in Fig. 4 while



(a) Basic access (no RTS/CTS).



(b) Mandatory RTS/CTS.

Fig. 6: The combined objective of Eq. (13) versus the number of visitor STAs, in the setting of Fig. 4, for proportionality weights $\alpha \in \{0,\, 0.5,\, 1\}$.

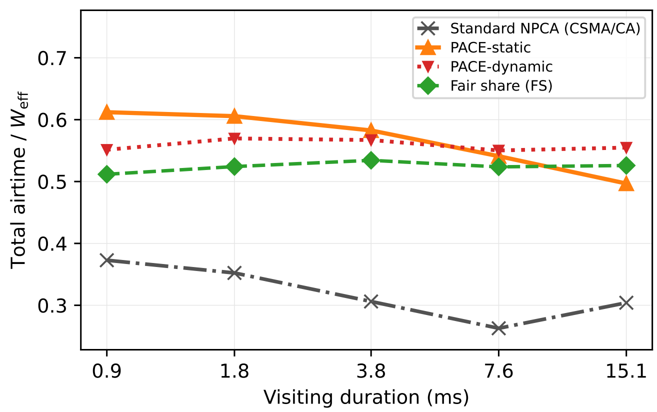
leaving the natives' share intact, since its gain comes from slots that contention had been wasting rather than from native airtime.

Efficiency and proportionality can also be read on one axis. Fig. 6 evaluates the objective of Eq. (13) across the visitor sweep for three settings of the weight α, which has no canonical value. At $\alpha = 0$ the objective reduces to log airtime and PACE sits at or above FS at every population. Raising α rewards proximity to the proportional line, which is FS's defining property, so FS moves ahead by a growing margin, up to 0.26 at $\alpha = 1$ under basic access, while PACE stays within about 0.1 of FS at the calibration weight $\alpha = 0.5$. The one reordering appears at $N_{\mathrm{vis}} = 5$ under the full weight $\alpha = 1$, where the standard baseline scores best of the three; a lightly loaded channel loses little airtime to its collisions and its share happens to sit near proportional, whereas FS structurally over-serves ($\rho \approx 1.6$) and PACE under-serves. From $N_{\mathrm{vis}} = 10$ onward, where contention actually stresses the window, no weight changes the ranking. PACE and FS remain close while the standard baseline trails by 0.3 to 1.1, its collapse in T dominating any proportionality credit. The calibration weight $\alpha = 0.5$ of Section IV-E thus sits in the middle of the range in which the comparison is insensitive to the weight. The comparison also flatters FS less than it seems. FS is a genie policy that transmits at $\tau^*(t) = 1/|\mathcal{V}(t)|$ with per-slot knowledge of the viable set, which presupposes not only the visitor population but the native one and every contender's frame fit, whereas PACE reaches these scores from local feedback alone, knowing neither N_{vis} nor N_{nat}. Near-parity with the genie at every weight is therefore the substantive result of the figure.

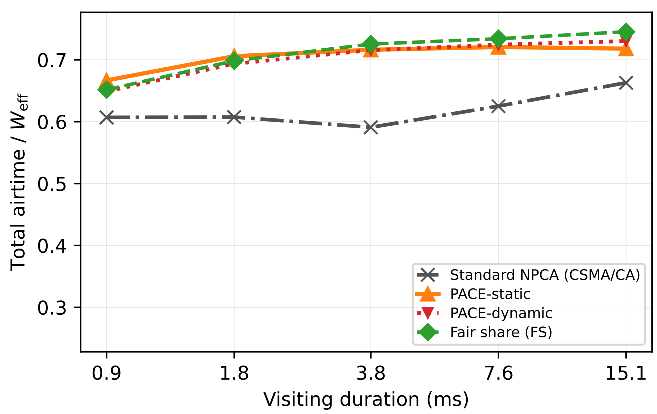
C. Impact of Visiting Duration

The window an NPCA visit receives is not a design choice but an environmental one. It is the remaining duration of the OBSS transmission that opened the opportunity, and it varies from sub-millisecond frames to transmission opportunity (TXOP)-scale bursts. Fig. 7 therefore sweeps the visiting duration from 0.9 to 15.1 ms, which corresponds to 1.6 to 27 mean visitor frames, at $N_{\mathrm{vis}} = 20$ visitors and $N_{\mathrm{nat}} = 10$ natives. This sweep is also where the window scaling of Section IV-E faces its test, so both variants appear, PACE-static with the fixed factor of Table II and PACE-dynamic with the factor recomputed from each visit's window, which falls from 2.76 at the shortest window to 1.28 at the longest.

Under basic access, shown in Fig. 7a, the standard visitors hold the channel at 0.26–0.37 across the sweep. However long the visit lasts, $\mathrm{CW}_{\min} = 16$ remains mismatched to the 30-station contention, and the synchronized collision cascades,

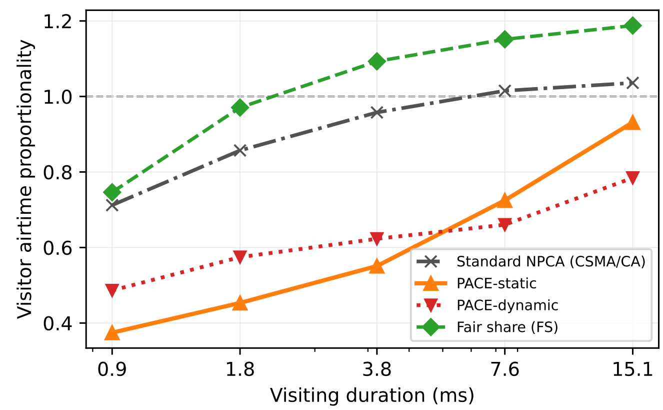


(a) Basic access (no RTS/CTS).

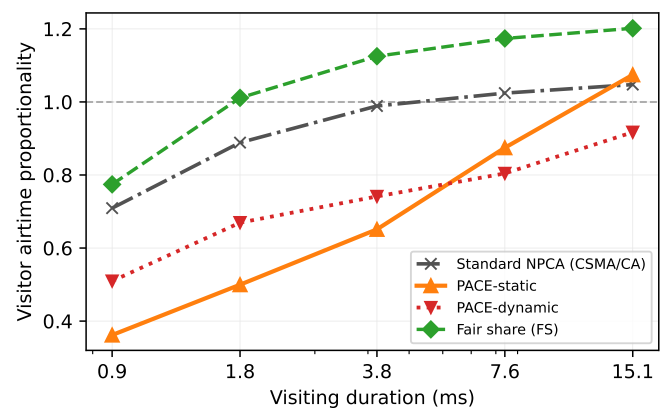


(b) Mandatory RTS/CTS.

Fig. 7: Total useful airtime versus the visiting duration (W_{eff} of 100–1680 slots, 0.9–15.1 ms) for the standard procedure, PACE-static, PACE-dynamic, and FS, with $N_{\text{vis}} = 20$ visitor STAs and $N_{\text{nat}} = 10$ natives; all other parameters as in Table II.



(a) Basic access (no RTS/CTS).



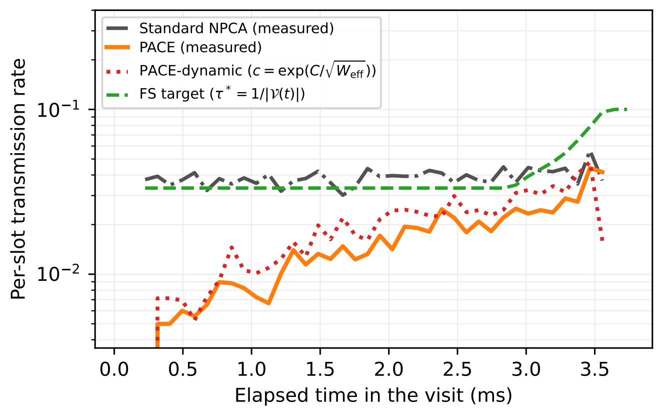
(b) Mandatory RTS/CTS.

Fig. 8: Visitor airtime proportionality ρ versus the visiting duration, in the setting of Fig. 7. The dashed line marks the proportional share $\rho = 1$.

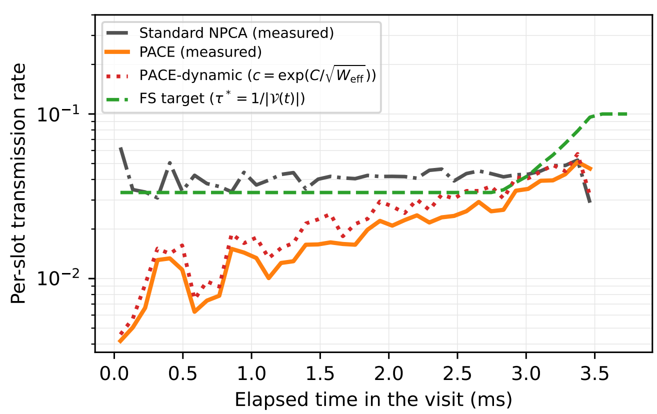
each charged a full frame length, consume a roughly constant fraction of whatever window is available. Both PACE variants deliver $1.5\times$ to $2.1\times$ the standard procedure, but they part at the ends of the sweep. PACE-static declines from 0.61 at 0.9 ms to 0.50 at 15.1 ms and slips below the FS line at the longest visits (0.50 against 0.53), since a factor calibrated at the reference window keeps raising the probability past the fair-share target once the visit lasts long enough for the overshoot to accumulate. PACE-dynamic holds 0.55–0.57 and stays above FS at every duration.

Fig. 8 shows what these totals cost in fairness. PACE-static spans ρ of 0.37–0.93 under basic access and 0.36–1.07 under RTS/CTS, so the same fixed factor leaves the visitors near a third of their proportional share in a 0.9-ms visit and pushes them past the proportional line in the longest protected ones. Its higher short-window total in Fig. 7a is earned the same way, by under-serving its own visitors while the natives fill the window. PACE-dynamic compresses the swing to 0.49–0.78 and 0.51–0.92 and stays below proportionality at every duration. A single coefficient thus errs in opposite directions at the two ends of the range, too timid where the visit is short and too aggressive where it is long, exactly the transient and fluctuation losses that Section IV-E balances, and the inverse-square-root rule corrects both without any additional signaling.

Under mandatory RTS/CTS, shown in Fig. 7b, the airtime differences narrow because the handshake caps the collision cost. The standard visitors stay 8–18% below PACE-static throughout, at 0.59–0.66, and the margin does not close as the window grows, since the losses caused by the backoff recur under saturation, where every completed exchange re-enters contention with a reset contention window. The two PACE variants track FS closely and nearly coincide in total airtime, with the long-window edge going to PACE-dynamic (0.73 against 0.72, FS at 0.75), so under this access mode the value of the window scaling lies less in airtime than in Fig. 8b, where PACE-dynamic keeps the visitors below the proportional line that the fixed factor crosses. At short durations both variants leave the visitors a smaller share of the window than the standard visitors do, the one-probe budget $\tau_0 = 1/W_{\text{eff}}$ leaves the feedback little room to ramp, but the composition is the fair one. The slots PACE declines to



(a) Basic access (no RTS/CTS).



(b) Mandatory RTS/CTS.

Fig. 9: Within-visit transmission dynamics in the reference setting of Fig. 4 ($N_{\text{vis}} = 20$, $N_{\text{nat}} = 10$, $W_{\text{eff}} = 420$ slots, 3.78 ms). The analytic FS target $\tau^*(t) = 1/|\mathcal{V}(t)|$ and the measured per-slot transmission rate of viable visitors under PACE-static, PACE-dynamic, and the standard procedure, averaged over 500 visits.

contest carry native traffic, keeping the total above the standard procedure at every duration.

D. Tracking the Time-Varying Fair-Share Target

The aggregate results of Sections V-A–V-C follow from the within-visit transmission behavior of each scheme, which Fig. 9 reports. Since the standard visitor holds no per-slot probability, we plot for every scheme the same model-free quantity, the measured fraction of viable visitors transmitting in each slot, against the analytic FS target. The target $\tau^*(t)$ stays near $1/(N_{\text{vis}}+N_{\text{nat}}) = 1/30$ for most of the visit and then rises steeply in the final slots, as PPDU-aware self-exclusion drains the viable set.

The standard-compliant visitor enters at $1/\text{CW}_{\min} = 1/16$, roughly twice the target, and its measured rate then remains near the target even though the scheme does not estimate it. The aggregate rate, however, conceals the failure mechanism. The countdown is deterministic and freezes during every busy period, so multiple backoff counters expire together once the medium becomes idle, and attempts at the same average rate collide far more often than independent ones, 84–86% of standard-visitor transmissions collide, against 59–60% for FS at a comparable rate (Table III). The aggregate rate is moreover sustained by the few STAs that happen to draw short backoffs, while each collision doubles the contention windows of the remaining STAs without an in-visit reset, so the airtime distribution across visitors becomes increasingly uneven although the aggregate rate is maintained.

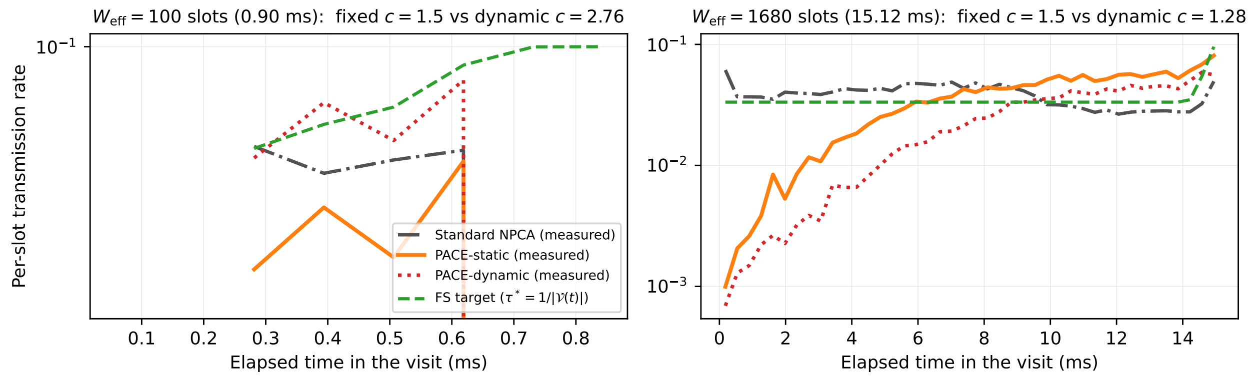
TABLE III: Per-Attempt Outcome of Visitor Transmissions (setting of Fig. 9, steady state)

Access	Scheme	Succ.	Coll.	Att./frame
basic	Standard	16%	84%	6.5
	PACE-static	44%	56%	2.3
	PACE-dynamic	40%	60%	2.5
	FS	40%	60%	2.5
RTS/CTS	Standard	14%	86%	7.3
	PACE-static	40%	60%	2.5
	PACE-dynamic	35%	65%	2.8
	FS	41%	59%	2.4

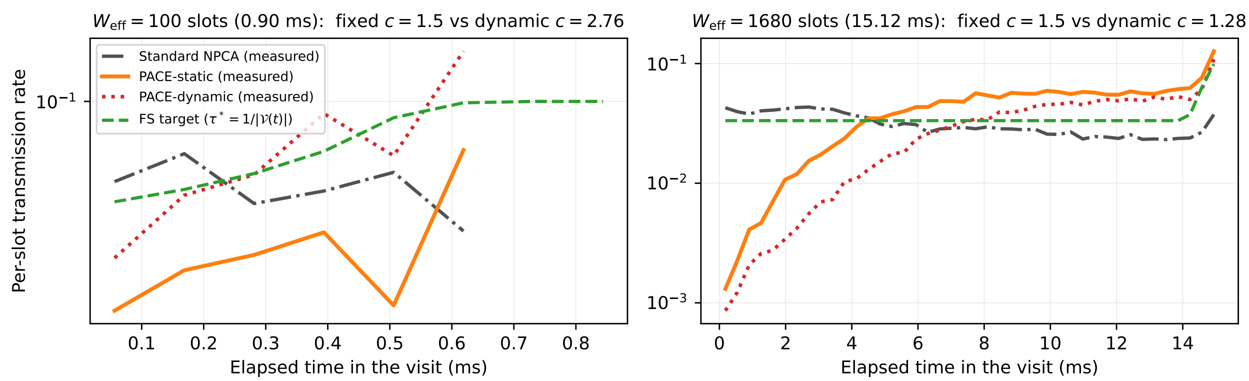
PACE requires neither knowledge of the target nor favorable backoff draws. It enters at $\tau_0 = 1/W_{\text{eff}}$, deliberately below the target, and the idle-driven multiplicative increase together with solo-copy propagation raises the measured rate toward the target from below over the course of the visit; being genuinely probabilistic, its attempts stay uncorrelated, and they collide at essentially the rate of the FS reference (Table III). Approaching the target from below is the safe direction established in Section IV-A. An under-aggressive probability costs a few idle slots, which the feedback recovers quickly, whereas over-aggressive or synchronized attempts consume the window in full-frame collisions that no subsequent adaptation can recover. The same qualitative picture holds in both access modes; under RTS/CTS the convergence is tighter because each corrective feedback event costs at most the constant L_{col} rather than a full frame.

Fig. 9 is drawn at the reference window, where the factor PACE-dynamic computes, 1.64, differs little from the fixed 1.5 and the two variants trace nearly the same path. The distinction appears at the ends of the window sweep, which Fig. 10 redraws. In a 0.9-ms visit the fixed factor never lifts the rate to the target before the window closes, which is the transient loss of Section IV-E made visible, whereas the per-visit factor of 2.76 arrives with slots to spare. In a 15.1-ms visit the same fixed factor crosses the target partway through and keeps the rate above it for the remainder, the accumulated fluctuation loss, while the factor of 1.28 settles onto the target and stays there. The two failure modes are symmetric, and correcting them at the trajectory level is what produces the airtime crossing and the compressed fairness swing of Figs. 7 and 8.

Table III classifies every visitor transmission attempt in the setting of Fig. 9 and summarizes the contention efficiency of each scheme. Only 14–16% of the standard visitors' transmission attempts succeed; the rest are synchronized collisions, so the standard procedure transmits six to seven times to deliver a single frame. PACE completes 40–44% of its attempts and



(a) Basic access (no RTS/CTS).



(b) Mandatory RTS/CTS.

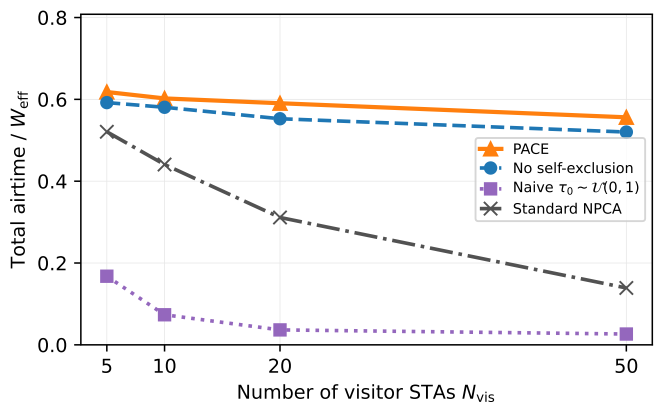
Fig. 10: Within-visit transmission dynamics at the two ends of the window sweep of Fig. 7 (W_{eff} of 100 and 1680 slots, 0.9 and 15.1 ms, $N_{\text{vis}} = 20$, $N_{\text{nat}} = 10$). PACE-static keeps the fixed factor $c = 1.5$ at both windows, while PACE-dynamic recomputes it per visit, $c = 2.76$ at the short window and $c = 1.28$ at the long one.

delivers a frame in about two and a half transmissions, which is essentially the per-attempt outcome of FS itself; PACE-dynamic, whose per-visit factor at this window is the slightly more aggressive 1.64, sits a few points lower at 35–40% while delivering the same airtime. A high success ratio alone could be achieved by transmitting rarely, so the result is meaningful only in combination with Fig. 4. PACE attains the highest total airtime of the three schemes while spending about the same number of transmissions per delivered frame as the fair-share reference, whereas the standard procedure retransmits the most and delivers the least; the scheme does not beat the fair operating point, it reaches it without being told where it is.

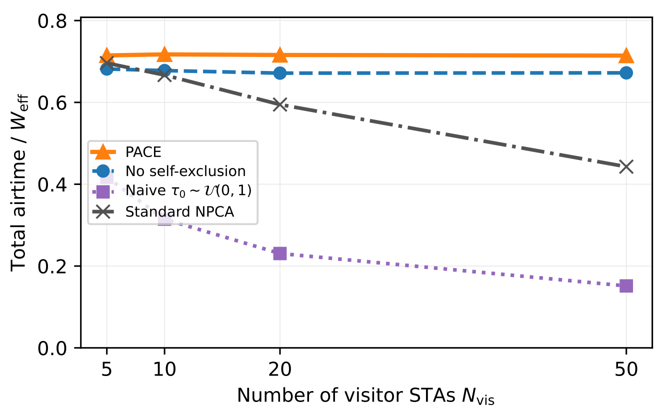
E. Ablation Study

PACE adds two mechanisms to the MIMD adaptation rule, the one-probe initialization $\tau_0 = 1/W_{\text{eff}}$ of Phase (a) and the PPDU-aware self-exclusion of Phase (b). Fig. 11 removes each in turn in the setting of Fig. 4. The initialization determines the behavior at the beginning of the visit and self-exclusion determines the behavior at its end. The third design choice, the adaptation step of Section IV-E, distinguishes itself only when the window varies, so its ablation is the static-to-dynamic comparison of Section V-C; at the reference window of this experiment the two variants nearly coincide, and the runs here use PACE-static.

Replacing the initialization with the knowledge-free choice $\tau_0 \sim \mathcal{U}(0, 1)$, a natural prior when no population estimate is available, reduces the total airtime under basic access to 0.17–0.03, below even the standard baseline at every population. The visit begins with a large number of simultaneous transmissions, each collision is charged the longest colliding frame, and the multiplicative decrease is too slow to bring τ to the operating point of order 10^{-2} before the window ends. Under RTS/CTS the totals of 0.42–0.15 also remain below the standard baseline, because a second failure mechanism operates in addition to the initial collisions. Visitors drawn near $\tau \approx 1$ transmit in almost every slot, so solo successes, the events that propagate a common operating point through the copy rule of (10), rarely occur, and the population cannot converge to a lower τ. The same experiment shows a fairness deficit as well. At $N_{\text{vis}} = 5$ under RTS/CTS the visitors occupy most of the useful airtime while the ten natives obtain almost none. A probabilistic access scheme without a deadline-scaled initialization therefore performs worse than the backoff procedure it replaces, so the conservative initialization is a precondition for PACE rather than an optimization.



(a) Basic access (no RTS/CTS).



(b) Mandatory RTS/CTS.

Fig. 11: Ablation study in the setting of Fig. 4: total useful airtime for full PACE, PACE without PPDU-aware self-exclusion, PACE without the one-probe initialization, and the standard-compliant baseline.

Removing self-exclusion causes a smaller but systematic loss that grows with the visitor population. Total airtime decreases by 3–7% across the sweep under both access modes, at $N_{\text{vis}} = 50$ from 0.556 to 0.520 under basic access and from 0.714 to 0.672 under RTS/CTS. Without Phase (b) a visitor whose frame no longer fits keeps contending and raises the collision rate observed by the STAs that can still succeed, and when it does obtain the channel it starts a transmission that the NPCA timer terminates, so the final part of the window is lost. Self-exclusion returns both losses to the remaining viable STAs. The ablated variant still exceeds the standard baseline by a wide margin, which indicates that the MIMD adaptation is the dominant mechanism and that self-exclusion prevents its gains from degrading near the deadline.

VI. Conclusion

This paper presented PACE, a probabilistic adaptive contention control scheme for non-primary channel access in IEEE 802.11bn. Inside the bounded NPCA visit the fair operating point $\tau^*(t) = 1/|\mathcal{V}(t)|$ rises toward the deadline, whereas BEB lowers its attempt rate after every collision and loses the window to synchronized collisions and frozen counters. PACE replaces the backoff countdown with a per-slot transmission probability governed by three local rules, namely the one-probe initialization $\tau_0 = 1/W_{\text{eff}}$, PPDU-aware self-exclusion, and MIMD adaptation with solo-copy consensus, and requires neither central coordination nor knowledge of the contending population.

Simulations under IEEE 802.11 timing with native DCF stations showed that PACE keeps the total useful airtime within 3% of a genie-aided fair-share reference under RTS/CTS-protected access, exceeds the standard-compliant baseline by up to 61% there and by up to $4.1\times$ under basic access, and holds the visitors' airtime at or below their population-proportional share across visit durations of one to fifteen milliseconds. Each delivered frame costs a quarter of the transmission attempts that the standard procedure requires. An ablation study confirmed that both the deadline-scaled initialization and the self-exclusion rule are necessary for these gains.

Future work includes a cost-aware decrease factor that weights collisions by their observed channel occupancy, partial-overlap topologies with spatial reuse, TXOP aggregation within a visit, and warm-starting τ across repeated visits to shorten the ramp-up in sub-millisecond windows.

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